

$$\xi^{\alpha\beta} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \xi_{\alpha\beta} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \sigma^{\mu}_{\alpha\dot{\alpha}} = (I, \vec{\sigma})$$

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$\Rightarrow g_{\mu\nu} \sigma^{\mu}_{\alpha\dot{\alpha}} \sigma^{\nu}_{\beta\dot{\beta}}$ must also be invariant

$$\sigma^{\mu}_{\alpha\dot{\alpha}} \sigma_{\mu\beta\dot{\beta}} = -2 \xi_{\alpha\beta} \xi_{\dot{\alpha}\dot{\beta}}$$

Similarly $\xi^{\alpha\beta} \xi^{\dot{\alpha}\dot{\beta}} \sigma^{\mu}_{\alpha\dot{\alpha}} \sigma^{\nu}_{\beta\dot{\beta}} = 2 g^{\mu\nu}$

* Generators $(S_L^{\mu\nu})_{\alpha}{}^{\beta}, (S_R^{\mu\nu})_{\dot{\alpha}}{}^{\dot{\beta}}$

$\rightarrow \xi_{\alpha\beta} = L(\Lambda)_{\alpha}{}^{\gamma} L(\Lambda)_{\beta}{}^{\delta} \xi_{\gamma\delta}$ (starting with)

Now, expand

$$\Lambda^{\mu}_{\nu} = \delta^{\mu}_{\nu} + \delta\omega^{\mu}_{\nu}$$

$$L(\Lambda)_{\alpha}{}^{\beta} = L(1 + \delta\omega) = \delta_a^b + \delta\omega^{\mu\nu} \frac{\partial}{\partial \omega^{\mu\nu}} L_{\alpha}{}^{\beta}$$

$$\Rightarrow \frac{\partial}{\partial \omega^{\mu\nu}} L_{\alpha}{}^{\beta} = \frac{i}{2} S_{\mu\nu}^{\alpha\beta}$$

$$\therefore L(1 + \delta)_{\alpha}{}^{\beta} = \delta_{\alpha}^{\beta} + \frac{i}{2} \delta\omega_{\mu\nu} (S_L^{\mu\nu})_{\alpha}{}^{\beta}$$

$\times \xi_{\gamma\delta}$

$$\xi_{\alpha\beta} = (\delta_{\alpha}^{\gamma} + \frac{i}{2} \delta\omega_{\mu\nu} (S_L^{\mu\nu})_{\alpha}{}^{\gamma}) (\delta_{\beta}^{\delta} + \frac{i}{2} \delta\omega_{\mu\nu} (S_L^{\mu\nu})_{\beta}{}^{\delta}) \xi_{\gamma\delta}$$

(RHS)

$$= (\delta_{\alpha}^{\gamma} \delta_{\beta}^{\delta} + \frac{i}{2} \delta\omega_{\mu\nu} (\delta_{\beta}^{\delta} (S_L^{\mu\nu})_{\alpha}{}^{\gamma} + \delta_{\alpha}^{\gamma} (S_L^{\mu\nu})_{\beta}{}^{\delta})) \xi_{\gamma\delta}$$

$$= \xi_{\alpha\beta} + \frac{i}{2} \delta\omega_{\mu\nu} (\xi_{\gamma\beta} (S_L^{\mu\nu})_{\alpha}{}^{\gamma} + \xi_{\alpha\delta} (S_L^{\mu\nu})_{\beta}{}^{\delta}) + \mathcal{O}(\omega^2)$$

$$0 = \frac{i}{2} \delta\omega_{\mu\nu} (-\xi_{\beta\delta} (S_L^{\mu\nu})_{\alpha}{}^{\gamma} + \xi_{\alpha\gamma} (S_L^{\mu\nu})_{\beta}{}^{\delta})$$

$$0 = \frac{i}{2} \delta\omega_{\mu\nu} (-(S_L^{\mu\nu})_{\beta\alpha} + (S_L^{\mu\nu})_{\alpha\beta})$$

and so

$$(S_L^{\mu\nu})_{\alpha\beta} = (S_L^{\mu\nu})_{\beta\alpha}$$

Likewise, we find

$$(S_R^{\mu\nu})_{\dot{\alpha}\dot{\beta}} = (S_R^{\mu\nu})_{\dot{\beta}\dot{\alpha}}$$

* The hermitz invariant object $\sigma^{\mu}_{\alpha\dot{\alpha}}$ will transform in the standard coordinate system

$$\sigma^{\mu}_{\alpha\dot{\alpha}} = \Lambda^{\mu}_{\nu} \frac{1}{\Lambda} \Lambda^{\beta} R(\Lambda)_{\dot{\beta}}{}^{\dot{\alpha}} \sigma^{\nu}_{\beta\dot{\beta}}$$

Substitute

$$\Delta^P{}_\tau = \delta^P{}_\tau + \frac{i}{2} \delta\omega_{\mu\nu} (S^{\mu\nu})^P{}_\tau$$

$$L(\Delta)_\alpha{}^B = \delta_\alpha^B + \frac{i}{2} \delta\omega_{\mu\nu} (S^{\mu\nu})_\alpha{}^B$$

$$R(\Delta)_{\dot{\alpha}}{}^{\dot{B}} = \delta_{\dot{\alpha}}^{\dot{B}} + \frac{i}{2} \delta\omega_{\mu\nu} (S^{\mu\nu})_{\dot{\alpha}}{}^{\dot{B}}$$

$$\Rightarrow (S^{\mu\nu})^P{}_\tau \equiv \frac{1}{i} (g^{\mu P} \delta^{\nu}_\tau - g^{\nu P} \delta^{\mu}_\tau)$$

Plugging this back in

$$\frac{1}{2} \delta\omega_{\mu\nu} (g^{\mu P} \delta^{\nu}_\tau - g^{\nu P} \delta^{\mu}_\tau) = \frac{1}{2} \delta\omega^P{}_\tau - \frac{1}{2} \delta\omega_\tau{}^P$$

$$\delta\omega_\tau{}^P = g^{\rho\sigma} \delta\omega_{\tau\sigma} = -g^{\rho\sigma} \delta\omega_{\sigma\tau} = -\delta\omega^P{}_\tau$$

$$= \delta\omega^P{}_\tau$$

as we would expect.

$$\begin{aligned} (RHS) &= (\delta_\tau^P + S_V^P{}_\tau) (\delta_\alpha^B + S_L^B{}_\alpha) (\delta_{\dot{\alpha}}^{\dot{B}} + S_R^{\dot{B}}{}_{\dot{\alpha}}) \sigma^\tau{}_{B\dot{B}} \\ &= (\delta_\tau^P + S_V^P{}_\tau) (\delta_\alpha^B \delta_{\dot{\alpha}}^{\dot{B}} + \delta_\alpha^{\dot{B}} S_L^B{}_{\dot{\alpha}} + \delta_\alpha^B S_R^{\dot{B}}{}_{\dot{\alpha}}) \\ &= \delta_\tau^P \delta_\alpha^B \delta_{\dot{\alpha}}^{\dot{B}} + \delta_\tau^P \delta_\alpha^{\dot{B}} S_L^B{}_{\dot{\alpha}} + \delta_\tau^P \delta_\alpha^B S_R^{\dot{B}}{}_{\dot{\alpha}} + \delta_\alpha^B \delta_{\dot{\alpha}}^{\dot{B}} S_V^P{}_\tau \\ &= \sigma^P{}_{B\dot{B}} + \frac{i}{2} \delta\omega_{\mu\nu} [\sigma^P{}_{B\dot{B}} (S^{\mu\nu})_\alpha{}^B + \sigma^P{}_{\alpha\dot{B}} (S^{\mu\nu})_{\dot{\alpha}}{}^{\dot{B}} + \sigma_{\alpha\dot{\alpha}}^\tau S_V^P{}_\tau] \end{aligned}$$

Thus $[] = 0$.

$$i\sigma^P{}_{B\dot{B}} (S_L)^{\mu\nu}{}_\alpha{}^B + i\sigma^P{}_{\alpha\dot{B}} (S_R)^{\mu\nu}{}_{\dot{\alpha}}{}^{\dot{B}} + \sigma_{\alpha\dot{\alpha}}^\tau (g^{\mu P} \delta^{\nu}_\tau - g^{\nu P} \delta^{\mu}_\tau) = 0$$

• Multiply by $\sigma_{\rho\delta\dot{\gamma}}$

$$\begin{aligned} i(-2\varepsilon_{B\gamma} \varepsilon_{\dot{\gamma}\dot{\delta}}) (S_L)^{\mu\nu}{}_\alpha{}^B + (-2\varepsilon_{\alpha\gamma} \varepsilon_{\dot{\gamma}\dot{\delta}}) (S_R)^{\mu\nu}{}_{\dot{\alpha}}{}^{\dot{B}} + \sigma_{\rho\gamma\dot{\gamma}} () &= 0 \\ 2i\varepsilon_{\alpha\gamma} (S_L)^{\mu\nu}{}_\alpha{}^\gamma + 2i\varepsilon_{\alpha\gamma} (S_R)^{\mu\nu}{}_{\dot{\alpha}}{}^{\dot{\gamma}} + () & \end{aligned}$$

$$\begin{aligned} () &= \sigma_{\rho\gamma\dot{\gamma}} \sigma_{\alpha\dot{\alpha}}^\tau (g^{\mu P} \delta^{\nu}_\tau - g^{\nu P} \delta^{\mu}_\tau) \\ &= \sigma_{\rho\gamma\dot{\gamma}}^\mu \sigma_{\alpha\dot{\alpha}}^\nu - \sigma_{\rho\gamma\dot{\gamma}}^\nu \sigma_{\alpha\dot{\alpha}}^\mu \end{aligned}$$

Thus

$$\sigma_{\rho\gamma\dot{\gamma}}^\mu \sigma_{\alpha\dot{\alpha}}^\nu - \sigma_{\rho\gamma\dot{\gamma}}^\nu \sigma_{\alpha\dot{\alpha}}^\mu + 2i\varepsilon_{\dot{\gamma}\dot{\delta}} (S_L)^{\mu\nu}{}_\alpha{}^\gamma + 2i\varepsilon_{\alpha\gamma} (S_R)^{\mu\nu}{}_{\dot{\alpha}}{}^{\dot{\delta}} = 0$$



Since S_L and S_R are symmetric in $(\alpha\gamma)$ and $(\dot{\alpha}\dot{\gamma})$

i) Multiply by $\epsilon^{\dot{\alpha}\dot{\beta}} : \epsilon^{\dot{\alpha}\dot{\beta}} (S_R)_{\dot{\alpha}\dot{\beta}} = 0$

$$\epsilon^{\dot{\alpha}\dot{\beta}} (\sigma^\mu_{\dot{\gamma}\dot{\delta}} \sigma^\nu_{\alpha\beta} - \sigma^\nu_{\dot{\gamma}\dot{\delta}} \sigma^\mu_{\alpha\beta}) + 2i \epsilon^{\alpha\beta} \epsilon_{\dot{\alpha}\dot{\beta}} (S_L^{\mu\nu})_{\alpha\beta}$$

$$\epsilon^{\alpha\beta} \epsilon_{\dot{\alpha}\dot{\beta}} = -2$$

$$\therefore (S_L^{\mu\nu})_{\alpha\beta} = \frac{1}{4i} \epsilon^{\dot{\alpha}\dot{\beta}} (\sigma^\mu_{\dot{\gamma}\dot{\delta}} \sigma^\nu_{\alpha\beta} - \sigma^\nu_{\dot{\gamma}\dot{\delta}} \sigma^\mu_{\alpha\beta})$$

or $(S_L^{\mu\nu})_{\alpha\beta} = \frac{i}{4} \epsilon^{\dot{\alpha}\dot{\beta}} (\sigma^\mu_{\alpha\dot{\beta}} \sigma^\nu_{\dot{\gamma}\beta} - \sigma^\nu_{\alpha\dot{\beta}} \sigma^\mu_{\dot{\gamma}\beta})$

~D If we instead multiply by $\epsilon^{\alpha\beta}$:

$$(S_R^{\mu\nu})_{\dot{\alpha}\dot{\beta}} = \frac{i}{4} \epsilon^{\alpha\beta} (\sigma^\mu_{\alpha\dot{\gamma}} \sigma^\nu_{\dot{\delta}\beta} - \sigma^\nu_{\alpha\dot{\gamma}} \sigma^\mu_{\dot{\delta}\beta})$$

Now, $\bar{\sigma}^{\mu\dot{\alpha}\alpha} \equiv \epsilon^{\dot{\alpha}\dot{\beta}} \epsilon^{\alpha\beta} \sigma^\mu_{\beta\dot{\beta}} = (1, -\vec{\sigma})$

i) $(S_L^{\mu\nu})_{\alpha\beta} \cdot \epsilon^{\beta\gamma}$

$$= i/4 \epsilon^{\beta\gamma} \epsilon^{\dot{\alpha}\dot{\beta}} (\sigma^\mu_{\alpha\dot{\beta}} \sigma^\nu_{\dot{\gamma}\beta} - \sigma^\nu_{\alpha\dot{\beta}} \sigma^\mu_{\dot{\gamma}\beta})$$

$$= (i/4) (\sigma^\mu_{\alpha\dot{\beta}} \bar{\sigma}^{\nu\dot{\alpha}\beta} - \sigma^\nu_{\alpha\dot{\beta}} \bar{\sigma}^{\mu\dot{\alpha}\beta})$$

$$(S_L^{\mu\nu})_{\alpha}{}^{\beta} = \frac{i}{4} (\sigma^\mu_{\alpha\dot{\beta}} \bar{\sigma}^{\nu\dot{\alpha}\beta} - \sigma^\nu_{\alpha\dot{\beta}} \bar{\sigma}^{\mu\dot{\alpha}\beta}) \quad (1/2, 0)$$

(Left handed Weyl Spinor Rep)

ii) $(S_R^{\mu\nu})_{\dot{\alpha}\dot{\beta}} \epsilon^{\dot{\beta}\dot{\gamma}}$

$$= i/4 \epsilon^{\dot{\alpha}\dot{\beta}} \epsilon^{\beta\gamma} (\sigma^\mu_{\alpha\dot{\beta}} \sigma^\nu_{\dot{\gamma}\beta} - \sigma^\nu_{\alpha\dot{\beta}} \sigma^\mu_{\dot{\gamma}\beta})$$

$$= (-i/4) (\bar{\sigma}^{\mu\dot{\beta}\alpha} \sigma^\nu_{\dot{\gamma}\beta} - \bar{\sigma}^{\nu\dot{\beta}\alpha} \sigma^\mu_{\dot{\gamma}\beta})$$

$$(S_R^{\mu\nu})^{\dot{\beta}}{}_{\dot{\alpha}} = -\frac{i}{4} (\bar{\sigma}^{\mu\dot{\beta}\alpha} \sigma^\nu_{\alpha\dot{\gamma}} - \bar{\sigma}^{\nu\dot{\beta}\alpha} \sigma^\mu_{\alpha\dot{\gamma}})$$

(Right handed Weyl Spinor Rep) $(0, 1/2)$

$(1/2, 0) \quad \psi_\alpha = \psi^\alpha \epsilon_\alpha$
 $(0, 1/2) \quad \bar{\psi}_{\dot{\alpha}} = \bar{\psi}^{\dot{\alpha}} \epsilon_{\dot{\alpha}}$ } Anti-commuting

$$\chi_\alpha(x) \psi_\beta(y) = -\psi_\beta(y) \chi_\alpha(x)$$

⇒ In this notation

$$\psi \chi = \psi^\alpha \chi_\alpha = \epsilon^{\alpha\beta} \psi_\beta \chi_\alpha = -\epsilon^{\alpha\beta} \chi_\alpha \psi_\beta = -\epsilon^{\beta\alpha} \chi_\alpha \psi_\beta = \chi^\beta \psi_\beta$$

$$\Rightarrow \boxed{\psi \chi = \chi \psi} \quad \psi^\dagger \chi^\dagger = \chi^\dagger \psi^\dagger$$

$$\rightarrow \bar{\psi} \bar{\sigma}^\mu \chi = \bar{\psi}_\alpha \bar{\sigma}^{\mu\dot{\alpha}\alpha} \chi_\alpha$$

This object behaves like a vector under Lorentz

$$U(\Lambda)^{-1} (\bar{\psi} \bar{\sigma}^\mu \chi)(y) U(\Lambda) = \Lambda^\mu_\nu [\bar{\psi} \bar{\sigma}^\nu \chi](\Lambda^{-1} x)$$

38.2 Show $(S_L^{\mu\nu})_a^b$ is consistent with the previous expressions

$$(S_L^{\mu\nu})_a^b = \frac{i}{4} (\sigma^\mu_{\alpha\dot{\gamma}} \bar{\sigma}^{\nu\dot{\gamma}\beta} - \sigma^\nu_{\alpha\dot{\gamma}} \bar{\sigma}^{\mu\dot{\gamma}\beta})$$

$$(S_L^{ij})_a^b = \frac{i}{4} (\sigma^i_{\alpha\dot{\gamma}} \bar{\sigma}^{j\dot{\gamma}\beta} - \sigma^j_{\alpha\dot{\gamma}} \bar{\sigma}^{i\dot{\gamma}\beta})$$

Since σ^i define an $SU(2)$ algebra

$$[\sigma^i, \sigma^j] = f^{ijk} \sigma_k \Rightarrow f^{ijk} = -2i \epsilon^{ijk}$$

$$\boxed{(S_L^{ij})_a^b = \frac{1}{2} \epsilon^{ijk} \sigma_k} \quad \text{as claimed}$$

$$\begin{aligned} (S_L^{0k})_a^b &= \frac{i}{4} (\sigma^0_{\alpha\dot{\gamma}} \bar{\sigma}^{k\dot{\gamma}\beta} - \sigma^k_{\alpha\dot{\gamma}} \bar{\sigma}^{0\dot{\gamma}\beta}) \\ &= \frac{i}{4} (\delta_{\alpha\dot{\gamma}} \bar{\sigma}^{k\dot{\gamma}\beta} - \sigma^k_{\alpha\dot{\gamma}} \delta^{\dot{\gamma}\beta}) \end{aligned}$$

$$\text{note } \bar{\sigma}^k = -\sigma^k \quad \text{and so} \\ = -i/2 \sigma^k$$

$$\text{Since } S^{\mu\nu} = -S^{\nu\mu}$$

$$\boxed{(S_L^{k0})_a^b = i/2 \sigma^k} \quad \text{as claimed}$$